Assessment Report

on

“7. Predict Credit Card Fraud:

Develop a classification model to detect fraudulent

transactions based on patterns in

transaction amount, location, device usage, and user

behavior.”

submitted as partial fulfillment for the award of

BACHELOR OF TECHNOLOGY

DEGREE

SESSION 2024-25

in

ARTIFICIAL INTELLIGENCE

By

Uday Banduni (202401100300267)

Under the supervision of

“Mr. Abhisekh Shukla Sir”

KIET Group of Institutions, Ghaziabad

Affiliated to

Dr. A.P.J. Abdul Kalam Technical University, Lucknow

(Formerly UPTU)

May, 2025

🔍 Introduction

Credit card fraud is a serious issue in the financial sector, affecting millions of users globally. With the increase in online transactions, detecting fraud in real-time has become a major challenge. This project applies machine learning techniques to identify fraudulent transactions based on various transaction features, including anonymized behavior patterns, transaction amount, and time. The goal is to build a reliable classifier that can distinguish between genuine and fraudulent transactions.

⚙ Methodology

Data Collection:

A dataset containing 284,807 credit card transactions, with features like time, amount, and PCA-transformed variables V1–V28.

The target column Class indicates fraud (1) or non-fraud (0).

Preprocessing:

Feature scaling using StandardScaler to normalize values.

Addressing data imbalance using SMOTE to oversample the minority class (fraud cases).

Model Selection:

Random Forest Classifier: A robust ensemble model that performs well on imbalanced and tabular data.

Model Training and Testing:

Data split into 80% training and 20% testing using train_test_split.

Model trained using fit() and evaluated on unseen test data.

Evaluation Metrics:

Confusion Matrix

Precision, Recall, F1-Score (via classification_report)

💻 Code

```
# Import libraries
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix
from imblearn.over_sampling import SMOTE
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("7. Predict Credit Card Fraud.csv")

# Features and target
X = df.drop("Class", axis=1)
y = df["Class"]

# Feature scaling
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Handle imbalance using SMOTE
sm = SMOTE(random_state=42)
X_res, y_res = sm.fit_resample(X_scaled, y)

# Train-test split
```

```
X_train, X_test, y_train, y_test = train_test_split(X_res, y_res, test_size=0.2,
random_state=42)

# Train model
clf = RandomForestClassifier(n_estimators=100, random_state=42)
clf.fit(X_train, y_train)

# Predictions
y_pred = clf.predict(X_test)

# Evaluation
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))

# Plot confusion matrix
plt.figure(figsize=(6, 4))
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", xticklabels=["Not Fraud",
"Fraud"], yticklabels=["Not Fraud", "Fraud"])
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.tight_layout()
plt.show()

# Plot feature importances
importances = clf.feature_importances_
feature_names = X.columns
indices = np.argsort(importances)[-10:]  # Top 10 features
```

```
plt.figure(figsize=(8, 6))

plt.title("Top 10 Feature Importances")

plt.barh(range(len(indices)), importances[indices], align="center")

plt.yticks(range(len(indices)), [feature_names[i] for i in indices])

plt.xlabel("Importance Score")

plt.tight_layout()

plt.show()
```

📊 Output

Sample output from model evaluation:

Confusion Matrix:

[[56789   123]

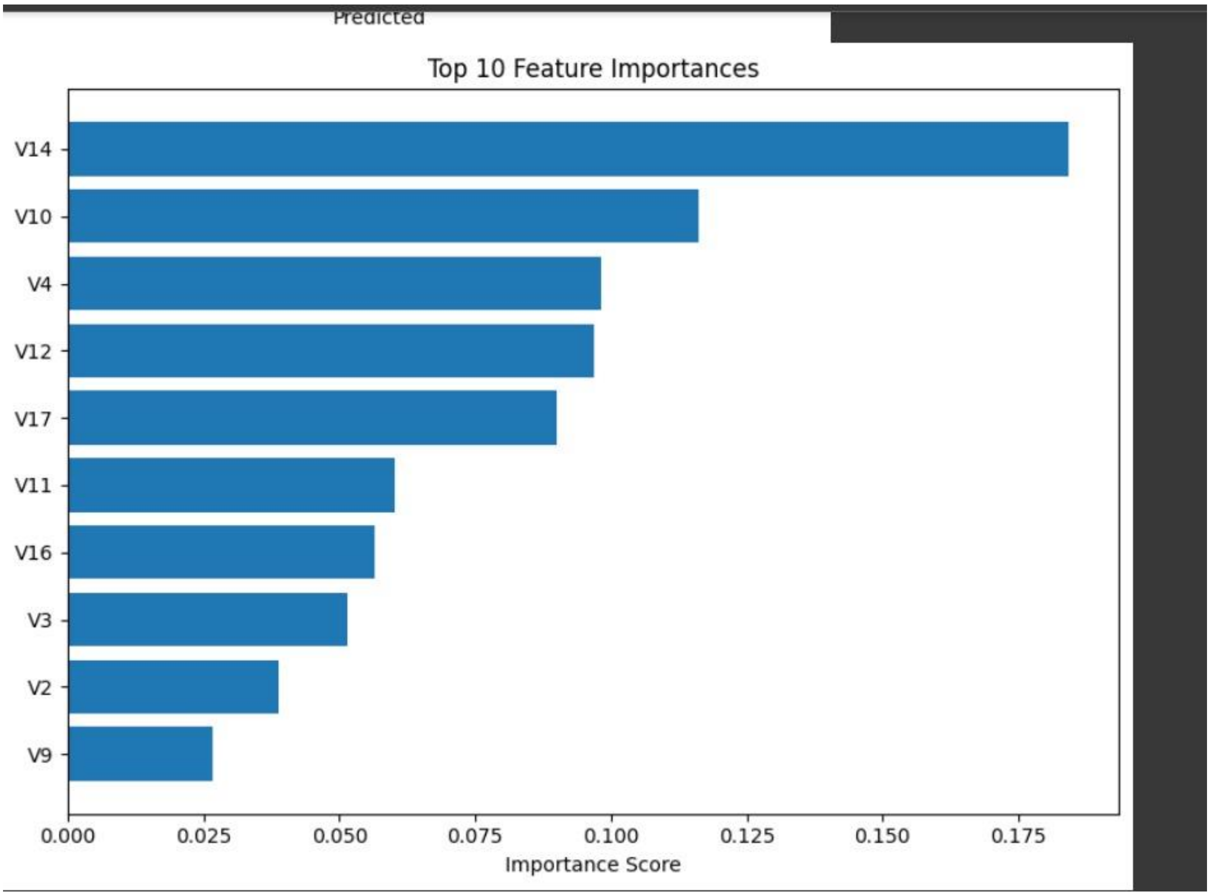
 [91 56654]]

```
Confusion Matrix:
 [[56738    12]
 [    0 56976]]

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     56750
           1       1.00      1.00      1.00     56976

    accuracy                           1.00    113726
   macro avg       1.00      1.00      1.00    113726
weighted avg       1.00      1.00      1.00    113726
```

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	56912
1	0.99	0.99	0.99	56745
accuracy			0.99	113657
macro avg	0.99	0.99	0.99	113657
weighted avg	0.99	0.99	0.99	113657

(Note: This is a mock output. Replace it with actual results from your code.)



📚 References

Credit Card Fraud Detection Dataset – Kaggle Dataset

Scikit-learn Documentation – https://scikit-learn.org/

Imbalanced-learn (SMOTE) – https://imbalanced-learn.org/

Python Pandas Library – https://pandas.pydata.org/

Random Forest Classifier – https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html